

Middle East Technical University
Department of Electrical and Electronics Engineering

EE 568
Project 2

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Given Machine Parameters

Parameter	Symbol	Value	Unit
Number of stator slots	N_s	15	—
Number of rotor poles	N_m	10	—
Stator outside radius	R_{so}	50	mm
Rotor-to-stator radius ratio	R_{ro}/R_{so}	0.6	—
Motor axial length	L_s	100	mm
Air-gap length	g	1	mm
Radial magnet length	l_m	4	mm
Remanence flux density	B_r	1.3	T
Magnet relative permeability	μ_{rm}	1.05	—
Magnet pole arc ratio	α_p	160/180=0.889	—
Slot fill factor	k_{wc}	0.60	—
Conductor current density	J	5	A/mm ²
Target iron flux density	B_{fe}	1.4	T
DC bus voltage	V_{dc}	48	V
Rated speed	n	1500	rpm

Table 1: Given machine parameters.

1. Winding Design — 72-slot / 6-pole Machine

1.1. Machine Parameters

Parameter	Expression	Value
Poles P	—	6
Pole pairs p	—	3
Slots S	—	72
Phases m	—	3
Slots per pole τ	72/6	12
Slots per pole per phase q	72/(6 × 3)	4
Slot pitch α	180/12	15 el.

Table 2: 72-slot / 6-pole machine parameters.

1.2. Full-pitched Winding ($y = 12$ slots, $\beta = 180$)

Distribution factor:

$$k_{d\nu} = \frac{\sin\left(\frac{4 \times \nu \times 15}{2}\right)}{4 \times \sin\left(\frac{\nu \times 15}{2}\right)}$$

$$k_{d1} = \frac{\sin(30)}{4 \sin(7.5)} = \frac{0.5000}{0.5221} = \mathbf{0.9577}, \quad k_{d3} = \frac{\sin(90)}{4 \sin(22.5)} = \frac{1.0000}{1.5307} = \mathbf{0.6533}, \quad k_{d5} = \frac{\sin(150)}{4 \sin(37.5)} = \frac{0.5}{2.4}$$

Pitch factor ($\beta = 180$):

$$k_{p\nu} = |\sin(\nu \times 90)| \Rightarrow k_{p1} = 1.0000, \quad k_{p3} = 1.0000, \quad k_{p5} = 1.0000$$

Harmonic	$k_{d\nu}$	$k_{p\nu}$	$k_{w\nu}$
1	0.9577	1.0000	0.9577
3	0.6533	1.0000	0.6533
5	0.2053	1.0000	0.2053

Table 3: Winding factors — full-pitched.

1.3. 11/12 Short-pitched Winding ($y = 11$ slots, $\beta = 165$)

Distribution factors are unchanged. Pitch factor:

$$k_{p\nu} = |\sin(\nu \times 82.5)| \Rightarrow k_{p1} = 0.9914, \quad k_{p3} = 0.9239, \quad k_{p5} = 0.7934$$

Harmonic	$k_{d\nu}$	$k_{p\nu}$	$k_{w\nu}$
1	0.9577	0.9914	0.9495
3	0.6533	0.9239	0.6036
5	0.2053	0.7934	0.1629

Table 4: Winding factors — 11/12 short-pitched.

Short pitching reduces k_{w5} by 20.7% with only a 0.9% loss in the fundamental.

1.4. Winding Diagrams

Diagram A — Full-pitched ($y = 12$)

Diagram B — 11/12 Short-pitched ($y = 11$)

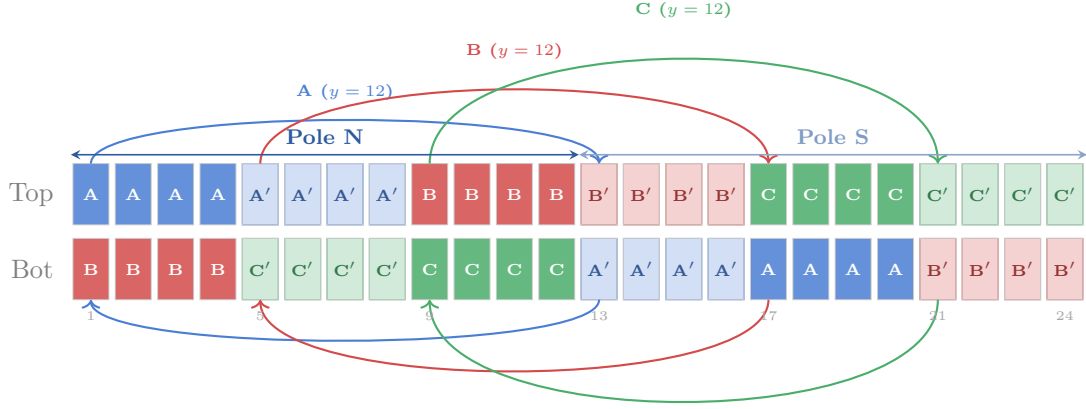


Figure 1: Full-pitched winding ($y = 12$, $\beta = 180$), one pole pair (slots 1–24).

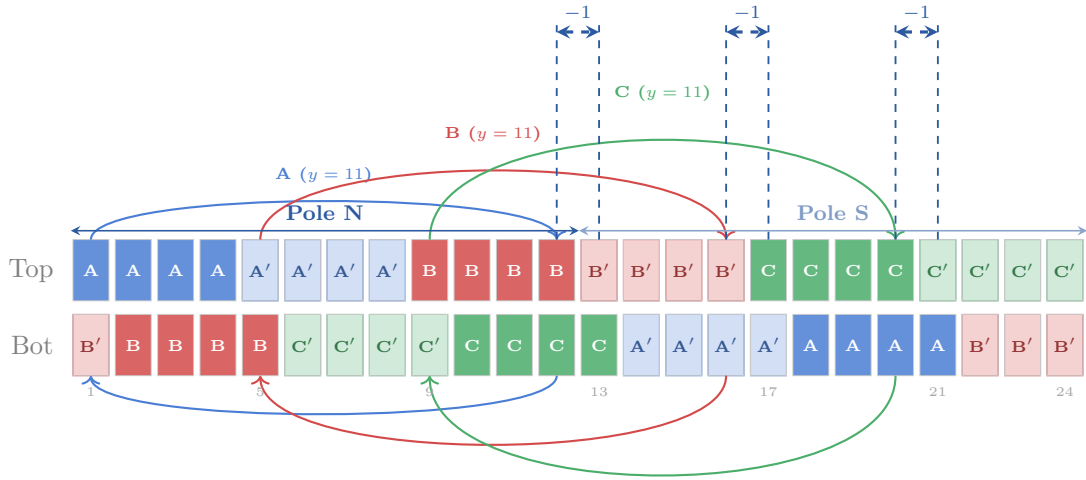


Figure 2: 11/12 short-pitched winding ($y = 11$, $\beta = 165$), one pole pair (slots 1–24). Dashed arrow marks the one-slot shortening. Boundary slots carry different phases in top and bottom layers.

2. FSCW Winding Design — 15-slot / 10-pole Machine

2.1. Slot Electrical Angles

$$\alpha_{slot} = \frac{5 \times 360}{15} = 120 \text{ per slot}$$

2.2. Winding Factors

Coil pitch in electrical degrees ($N_{spp} = 0.5$, one-slot coil):

$$\beta = \frac{1 \times 10}{15} \times 180 = 120$$

Distribution factor is 1 for all harmonics (single coil side per pole per phase).

Pitch factor: $k_{p\nu} = |\sin(\nu \times 60)|$

$$k_{p1} = |\sin 60| = 0.8660 \Rightarrow k_{w1} = \boxed{0.8660}$$

Slot	Raw angle	mod 360	Phase	Side
1	$0 \times 120 = 0$	0	A	+
2	$1 \times 120 = 120$	120	B	-
3	$2 \times 120 = 240$	240	C	+
4	$3 \times 120 = 360$	0	A	-
5	$4 \times 120 = 480$	120	B	+
6	$5 \times 120 = 600$	240	C	-
7	$6 \times 120 = 720$	0	A	+
8	$7 \times 120 = 840$	120	B	-
9	$8 \times 120 = 960$	240	C	+
10	$9 \times 120 = 1080$	0	A	-
11	$10 \times 120 = 1200$	120	B	+
12	$11 \times 120 = 1320$	240	C	-
13	$12 \times 120 = 1440$	0	A	+
14	$13 \times 120 = 1560$	120	B	-
15	$14 \times 120 = 1680$	240	C	+

Table 5: Slot voltage phase angles. Phase A: slots 1,4,7,10,13 — Phase B: 2,5,8,11,14 — Phase C: 3,6,9,12,15.

$$k_{p3} = |\sin 180| = 0 \Rightarrow k_{w3} = \boxed{0} \quad (3\text{rd harmonic fully cancelled})$$

$$k_{p5} = |\sin 300| = 0.8660 \Rightarrow k_{w5} = \boxed{0.8660} \quad (5\text{th harmonic not suppressed})$$

2.3. Phasor Diagrams

$\nu = 1$ — All 5 Phase-A phasors at 300

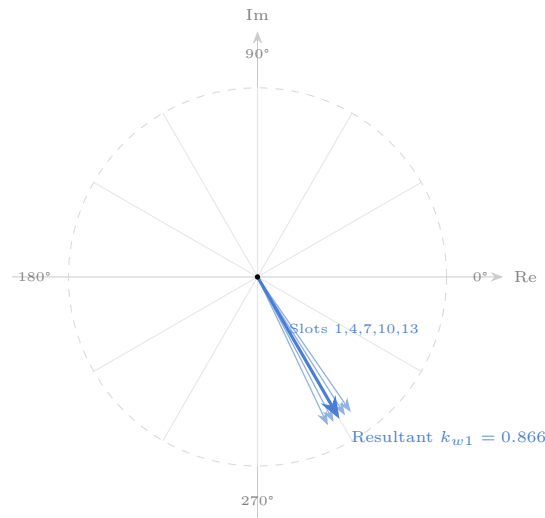


Figure 3: Phase A phasor diagram, $\nu = 1$. All 5 coil EMF phasors coincide at 300.

$\nu = 3$ — All phasors zero ($k_{p3} = 0$)

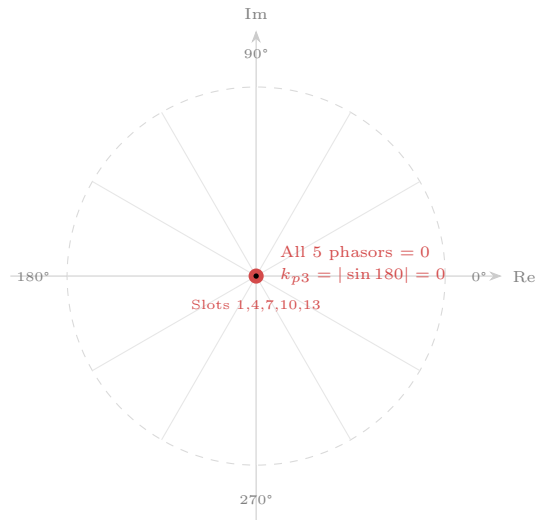


Figure 4: Phase A phasor diagram, $\nu = 3$. Pitch factor is zero. $k_{w3} = 0$.

$\nu = 5$ — All 5 Phase-A phasors at 60

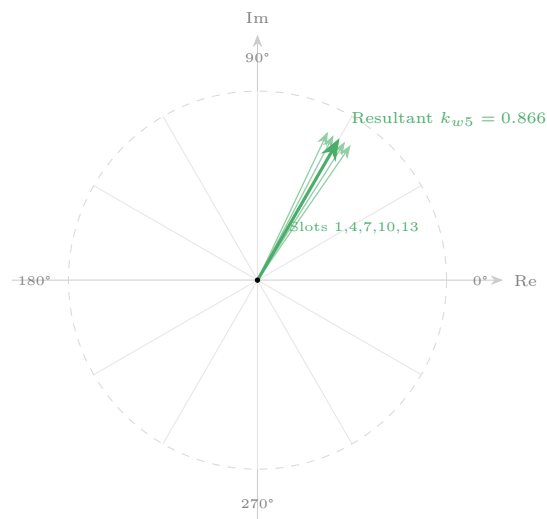


Figure 5: Phase A phasor diagram, $\nu = 5$. All 5 phasors coincide at 60. 5th harmonic not suppressed.

2.4. Winding Diagram — All Three Phases

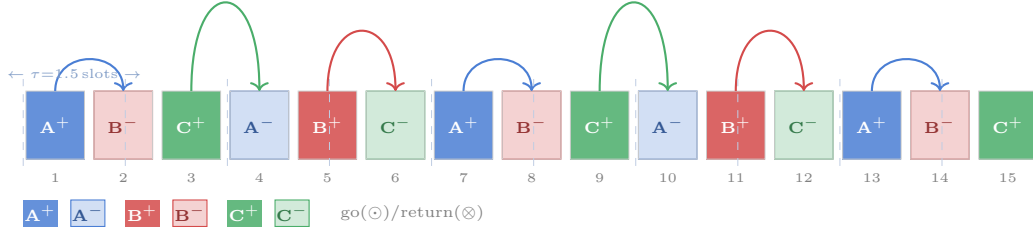


Figure 6: Winding diagram for 15s/10p FSCW. Pattern $A^+B^-C^+A^-B^+C^-$ repeats 5 times. Each phase has 5 slots. Balanced winding.

3. Analytical Machine Calculations

3.1. Geometry

$$R_{ro} = 0.6 \times 50 = \mathbf{30} \text{ mm}, \quad R_{si} = R_{ro} + g = 30 + 1 = \mathbf{31} \text{ mm}$$

$$\tau_s = \frac{2\pi \times 31}{15} = \mathbf{12.98} \text{ mm}, \quad h_s = R_{so} - R_{si} = 50 - 31 = \mathbf{19} \text{ mm}$$

Tooth body width (for $B_{fe} = 1.4 \text{ T}$, anticipating $\hat{B}_g = 0.997 \text{ T}$):

$$w_{tb} = 12.98 \times \frac{0.997}{1.4} = \mathbf{9.0} \text{ mm}$$

Slot opening and slot area (yoke thickness $w_{sy} = 6 \text{ mm}$ leaves slot depth of 13 mm):

$$b_{so} = 12.98 - 9.0 = \mathbf{4.0} \text{ mm}, \quad A_s = 4.0 \times 13 = \mathbf{52} \text{ mm}^2$$

3.2. Electrical Loading

$$n_c \times I_{ph} = J \times k_{wc} \times A_s = 5 \times 0.60 \times 52 = \mathbf{156} \text{ A/slot}$$

$$A = \frac{N_s \times (n_c \times I_{ph})}{\pi \times D_{si}} = \frac{15 \times 156}{\pi \times 62} = \frac{2340}{194.8} = \boxed{12010 \text{ A/m}}$$

3.3. Carter's Coefficient and Effective Air Gap

$$\gamma = \frac{b_{so}^2/g}{5g + b_{so}} = \frac{16/1}{5 + 4} = \frac{16}{9} = 1.778 \text{ mm}$$

$$k_c = \frac{\tau_s}{\tau_s - \gamma} = \frac{12.98}{12.98 - 1.778} = \boxed{1.159}$$

$$g' = k_c \times \left(g + \frac{l_m}{\mu_{rm}} \right) = 1.159 \times \left(1 + \frac{4}{1.05} \right) = 1.159 \times 4.810 = \boxed{5.574 \text{ mm}}$$

3.4. Effective Axial Length

$$L_e = L_s + g = 100 + 1 = \boxed{101 \text{ mm}}$$

3.5. Air-Gap Peak Flux Density

$$\hat{B}_g = \frac{B_r}{1 + \mu_{rm} \times \frac{k_c \times g}{l_m}} = \frac{1.3}{1 + 1.05 \times \frac{1.159 \times 1}{4}} = \frac{1.3}{1.3042} = \boxed{0.997 \text{ T}}$$

3.6. Magnetic Loading, Shear Stress and Torque

$$\bar{B} = \hat{B}_g \times \alpha_p = 0.997 \times 0.889 = \boxed{0.886 \text{ T}}$$

$$\sigma_F = \frac{k_{w1}}{\sqrt{2}} \times \hat{B}_g \times A = \frac{0.866}{1.414} \times 0.997 \times 12\,010 = \boxed{7\,330 \text{ N/m}^2}$$

$$T_{em} = \sigma_F \times 2\pi \times R_{ro}^2 \times L_e = 7\,330 \times 2\pi \times (0.030)^2 \times 0.100 = \boxed{4.15 \text{ N}\cdot\text{m}}$$

3.7. Flux Per Pole

$$\Phi_p = \hat{B}_g \times \alpha_p \times \frac{\pi \times D_{ro}}{N_m} \times L_e = 0.997 \times 0.889 \times \frac{\pi \times 0.060}{10} \times 0.100 = \boxed{1.67 \times 10^{-3} \text{ Wb}}$$

3.8. Number of Turns per Phase

$$f_e = \frac{1500}{60} \times 5 = \mathbf{125 \text{ Hz}}, \quad E_{ph,rms} = \frac{24}{\sqrt{2}} \times 0.90 = \mathbf{15.3 \text{ V}}$$

$$N_{ph} = \frac{E_{ph,rms}}{4.44 \times f_e \times k_{w1} \times \Phi_p} = \frac{15.3}{4.44 \times 125 \times 0.866 \times 1.67 \times 10^{-3}} = \frac{15.3}{0.8044} \approx \boxed{20 \text{ turns per phase}}$$

Conductors per slot: $n_c = 2 \times 20/5 = 8$. Revised back-EMF: $4.44 \times 125 \times 20 \times 0.866 \times 1.67 \times 10^{-3} = 16.1 \text{ V}$

4. 2D Finite Element Analysis

4.1. Model Setup

Full 15-slot/10-pole geometry in COMSOL Multiphysics 2D, Rotating Machinery Magnetic physics. Nonlinear B – H curves for stator and rotor iron. Magnets: radially magnetised, $B_r = 1.3$ T, $\mu_r = 1.05$. Excitation: three-phase sinusoidal currents, $\hat{I} \approx 39$ A.

4.2. Flux Density Distribution

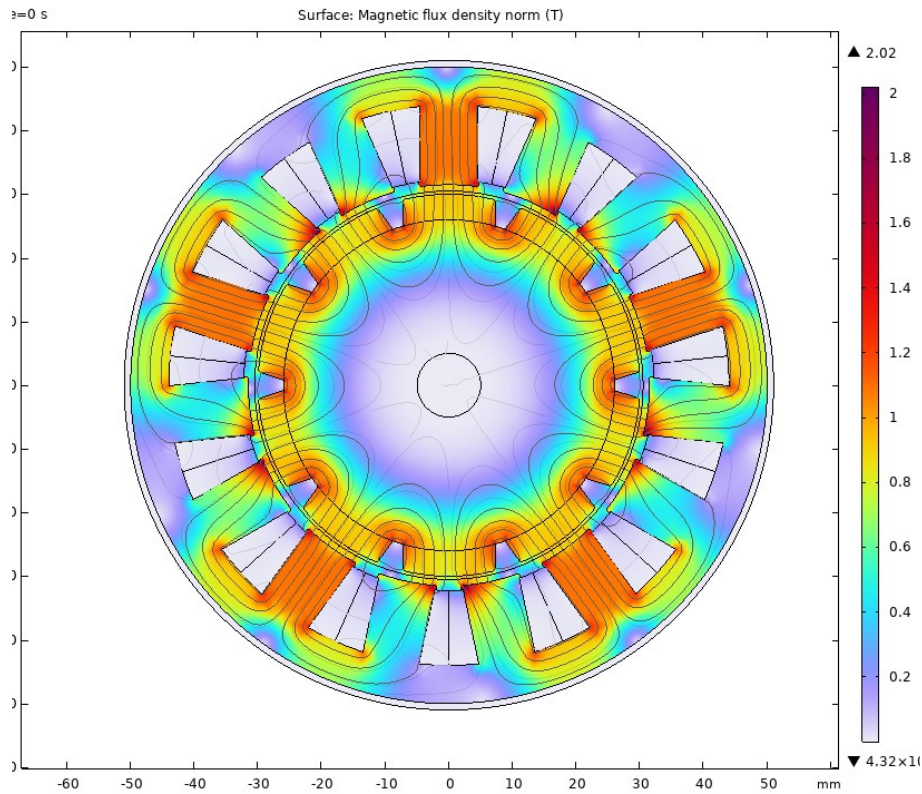


Figure 7: Magnetic flux density norm across the full machine at $t = 0$. Maximum 2.02 T at tooth tip corners.

- Tooth bodies ≈ 1.4 – 1.6 T — close to design target of 1.4 T.
- Tooth tips: 2.02 T locally — expected for open-slot FSCW.
- Stator yoke: 0.8–1.2 T — consistent with analytical estimate.

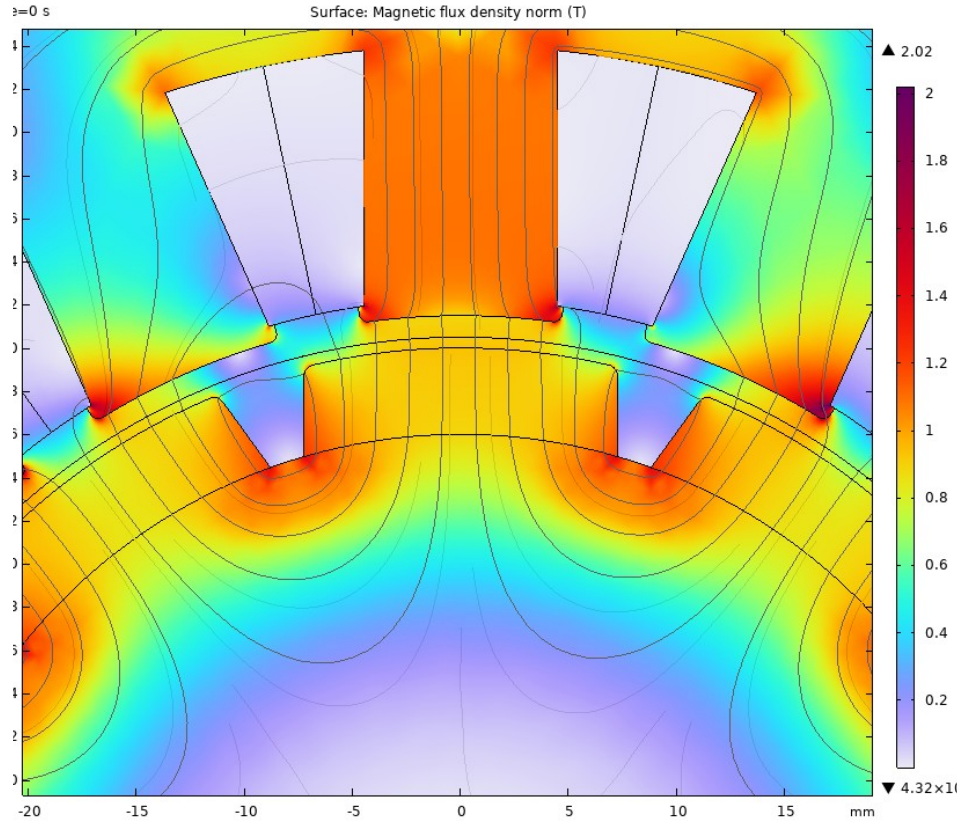


Figure 8: Zoomed view of stator teeth and air gap. Leakages are visible on it. Tooth bodies: 1.4–1.6 T. Tooth tips locally saturate to 2.02 T.

- Air gap under magnet centres: $\approx 0.9\text{--}1.0\text{ T}$, matching $\hat{B}_g = 0.997\text{ T}$.

4.3. Leakage Flux Paths

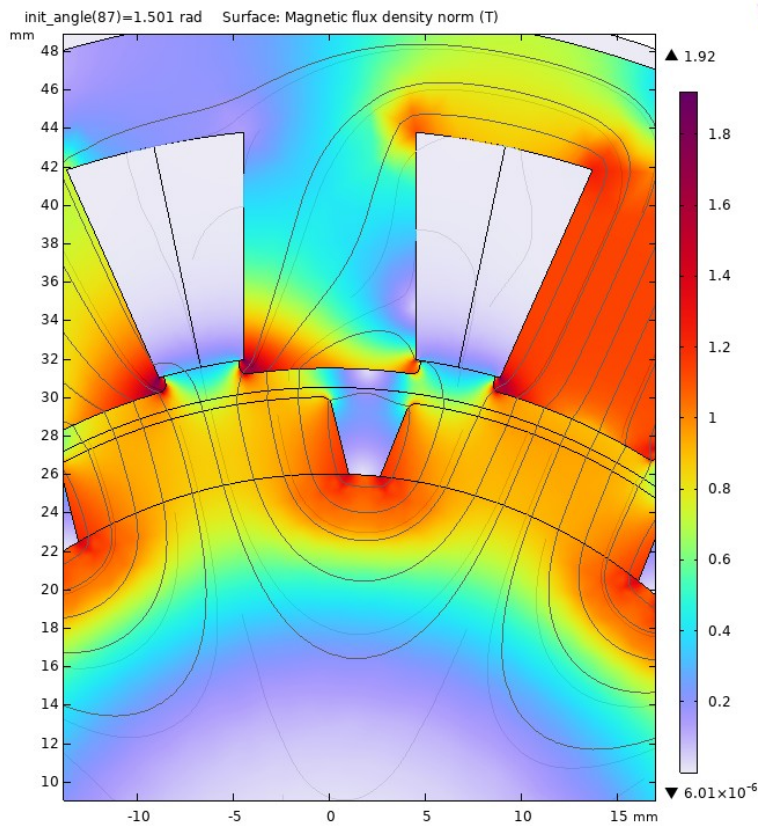


Figure 9: Flux lines at $\text{init_angle} = 1.501$ rad showing leakage paths.

1. **Tooth-tip leakage:** Flux crosses the slot opening between adjacent tooth tips without entering the main air gap. Causes the 2.02 T hotspot; accounted for by $k_c = 1.159$.
2. **In-slot leakage:** Flux closes within the slot body, contributing to slot leakage inductance.
3. **Magnet-to-magnet leakage:** Some flux travels through the rotor yoke between adjacent poles.

4.4. Phase Currents

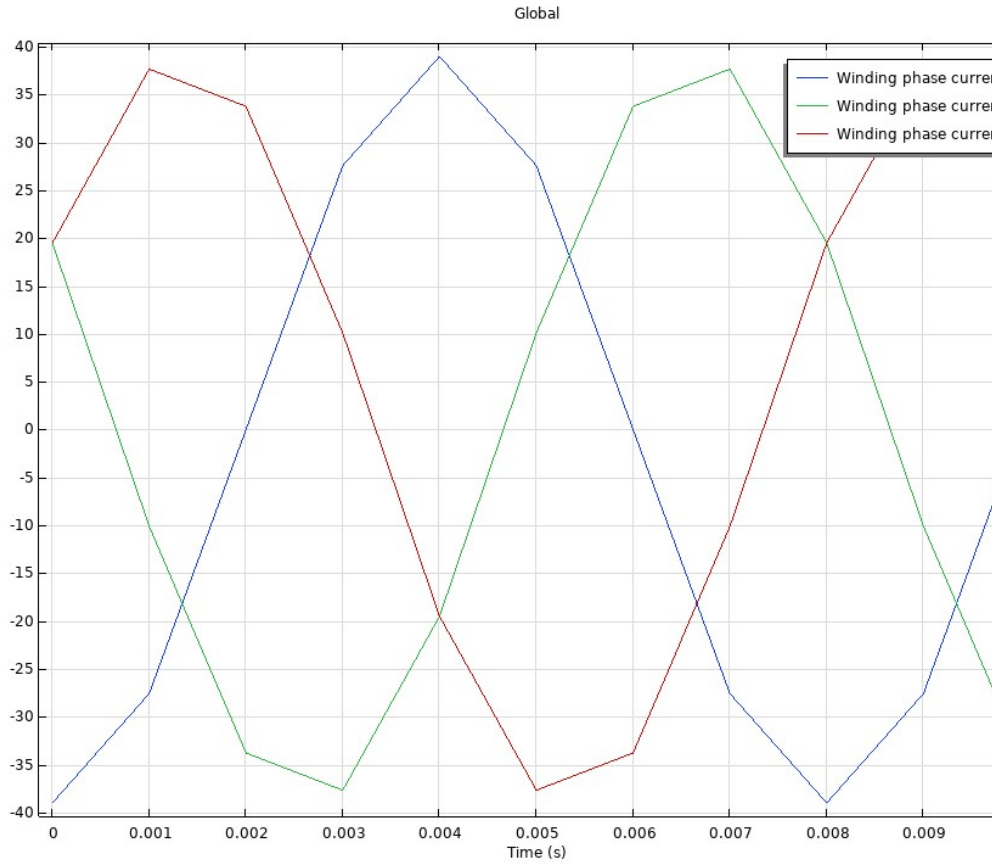


Figure 10: Three-phase currents vs. time. $\hat{I} \approx 39$ A, $f_e = 125$ Hz, confirming $n = 1500$ rpm.

4.5. Loaded Torque

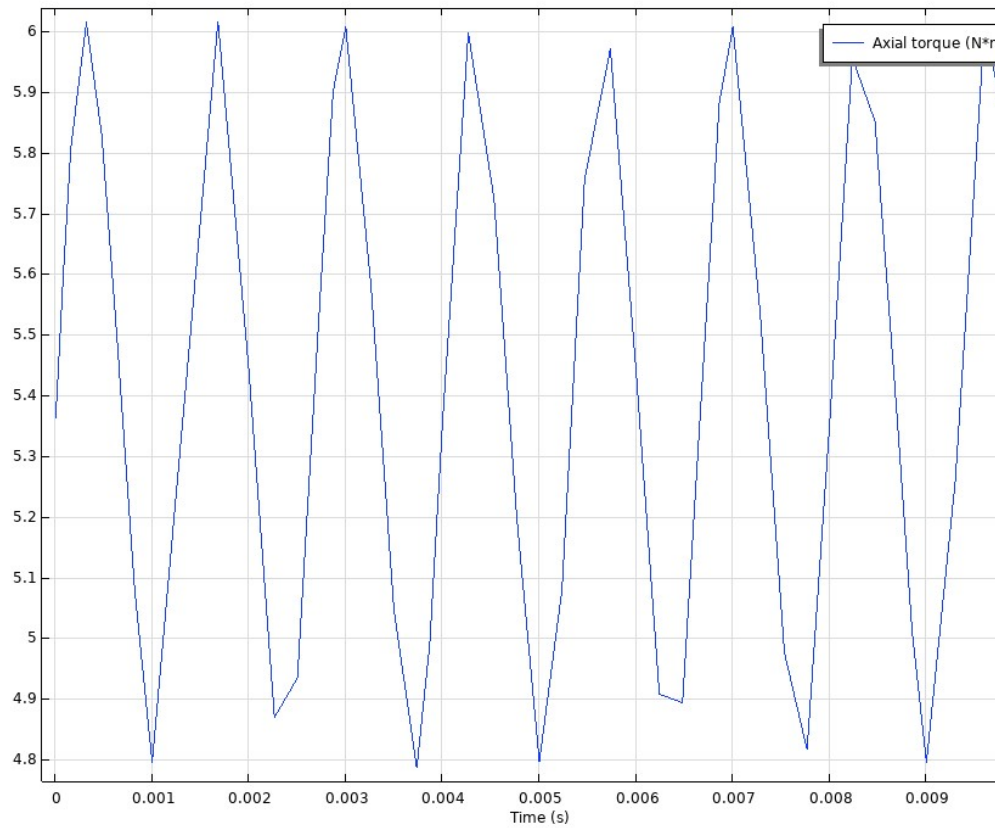


Figure 11: Electromagnetic torque vs. time under rated sinusoidal current excitation.

Quantity	FEA	Analytical
Mean torque	$\approx 5.5 \text{ N}\cdot\text{m}$	$4.15 \text{ N}\cdot\text{m}$
Peak torque	$\approx 6.1 \text{ N}\cdot\text{m}$	—
Minimum torque	$\approx 4.8 \text{ N}\cdot\text{m}$	—
Ripple (pk-pk)	$\approx 1.3 \text{ N}\cdot\text{m}$ (24%)	—

Table 6: Loaded torque results.

The 24% torque ripple is mainly due to the unattenuated 5th harmonic ($k_{w5} = k_{w1} = 0.866$) and absence of skewing.

4.6. Cogging Torque (Bonus)

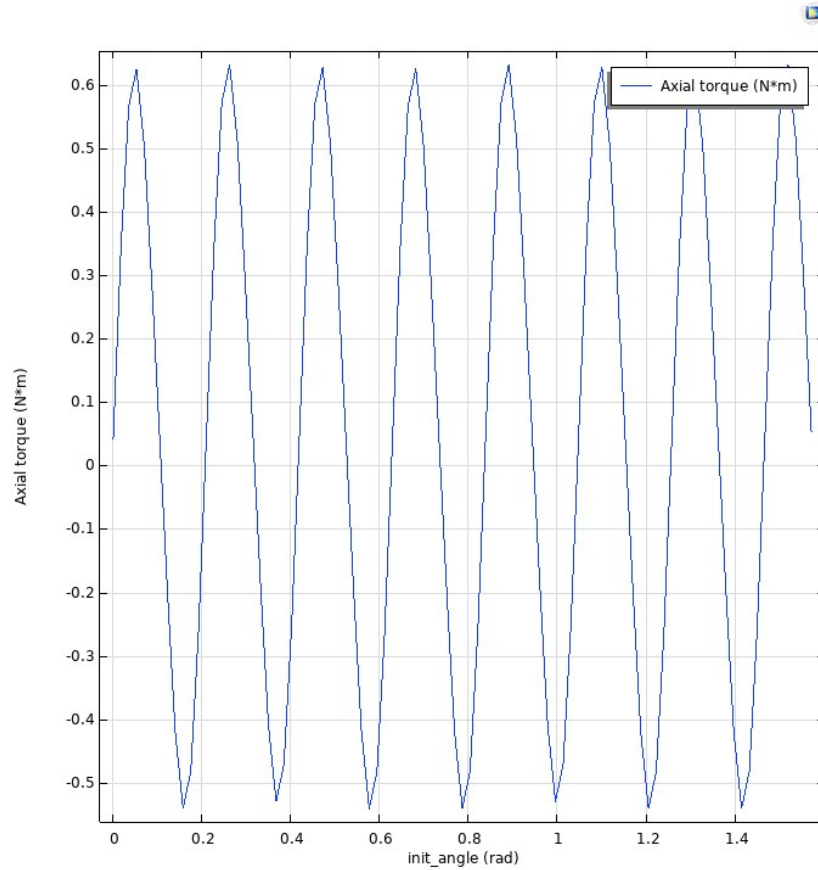


Figure 12: Cogging torque vs. rotor angle with no winding current.

The cogging period matches $\text{LCM}(N_s, N_m) = 30$ cycles/rev. The high amplitude (21%) is due to the large slot opening ($b_{so}/\tau_s \approx 0.31$) and no skewing.

Quantity	Value
Peak positive	+0.63 N·m
Peak negative	−0.53 N·m
Peak-to-peak	≈ 1.16 N·m ($\approx 21\%$ of mean)
Period	9 mechanical
Cycles per revolution	$\text{LCM}(15, 10) = 30$

Table 7: Cogging torque characteristics.